



BAHIR DAR UNIVERSITY
INSTITUTES OF TECHNOLOGY
FACULTY OF COMPUTING
COMPUTER SCIENCE PROGRAM

1. General Information of the course

Course Name	Computer Vision And Image Processing
Course Code	Cosc4102
Module Name	Intelligent System
Status Of The Course	Elective
Per-Request	None
Mode Of Delivery	Semester wise
Contact Hours (Per Week)	2 lecture , 3 lab
Instructor Information	Name :Alebachew M. Email : aleba.m.d@gmail.com Office :Tana building office No 310

2. Course description

This is an introductory course in processing Grey-scale and color images. It will introduce students to the basic principles of processing digital signals and how those principles apply to images. These fundamentals will include sampling theory, transforms in appearance and geometry, filtering and object segmentation. The course will also cover a series of basic image processing problems including enhancement, reconstruction, segmentation, feature detection, and compression.

3. Course objective

At the end of this course, students are expected to:

- ▶ Define fundamental concepts of computer vision and image processing system
- ▶ Apply pre-processing operations in image enhancement
- ▶ Analyze 2-dimensional signals in the frequency domain through the Fourier transform
- ▶ Describe the basic concepts of color image processing
- ▶ Apply image restoration, reconstruction and compression operations
- ▶ Design and implement image processing algorithm with Python.

4. Course Learning Outcomes (CLOs)

CLO1. Explain the fundamental concepts, terminology, and components of computer vision and digital image processing systems, including image formation and representation.

CLO2. Represent and analyze digital images using appropriate mathematical models, sampling, and quantization techniques.

CLO3. Apply spatial domain image processing techniques such as intensity transformations, histogram processing, and spatial filtering for image enhancement.

CLO4. Analyze and process images in the frequency domain using Fourier series and Fourier transform techniques for image enhancement and filtering.

CLO5. Identify and model image degradation and noise, and apply appropriate image restoration and reconstruction techniques.

CLO6. Explain the principles of image compression, data redundancy, and information theory, and apply basic compression techniques such as Huffman, arithmetic, and dictionary-based coding.

CLO7. Describe color fundamentals and color models, and apply color image processing and morphological operations for image analysis tasks.

CLO8. Implement image segmentation, edge detection, and feature extraction techniques for object recognition applications.

CLO9. Design and implement basic image processing and computer vision algorithms using Python or MATLAB to solve practical problems.

CLO10. Apply introductory machine learning and deep learning approaches for object recognition and three-dimensional image analysis

5. COURSE TIME SCHEDULE AND TEACHING METHODOLOGY

Course content	Time schedule	CLO
Chapter 1. Introduction to computer vision and image processing 1.1 What is a computer vision? 1.2 What is image? 1.3 Related fields in CV 1.4 Computer Vision Vs image processing 1.5 Application of CV and IP 1.6 Different Image processing examples 1.7 Fundamental steps in image processing	week 1, 2	CLO1
Chapter 2: Digital image fundamentals 2.1 Basic concept of image 2.2 Digital image Representation	Weeks 3, 4	CLO2

2.3 Digital image acquisition process 2.4 Image sampling and quantization 2.5 Representation of different image type's 2.6 Mathematical Tools used in Digital Image Processing		
Chapter 3: Spatial Domain Image Processing 3.1 Spatial Processing of Digital Images 3.2 Basic Intensity Transformation Functions 3.3 Histogram of images 3.4 Spatial filtering 3.5 Histogram processing 3.6 Combining Spatial Enhancement Methods	Weeks 5	CLO3 , CLO9
Mid exam	Week 8	CLO 1,2,3
Chapter 4: Frequency Domain Image Processing 4.1 Introduction to frequency domain 4.2 Frequency spectra 4.3 Fourier series 4.4 Continuous One-Dimensional Fourier Transform and Its Inverse 4.5 Discrete One-Dimensional Fourier Transform and Its Inverse 4.6 Frequency information of images 4.7 Image enhancement in frequency domain	Weeks 6,7	CLO4, CLO9
Chapter 5: Image Restoration and Reconstruction 5.1 Introduction 5.2 Models of images and noise 5.3 Estimation of noise parameters 5.4 Restoration in the presence of noise	Weeks 8,9	CLO5, CLO9
Chapter 6: Image Compression 6.1 Basic definition of image compression 6.2 Data redundancy 6.3 Elements of information theory 6.4 General mechanism and types of data compression and image restoration 6.5 Huffman coding 6.6 Arithmetic coding 6.7 Dictionary based coding 6.8 Bit-plane coding	Weeks 10,11	CLO6, CLO9
Chapter 7: Color Image Processing 7.1 Color Fundamentals 7.2 Color Models (color space or color system) 7.3 Basics of Full-Color Image Processing 7.4 Morphological image processing 7.5 Color Transformations	Weeks 12,13	CLO7, CLO9
Chapter 8: Object Recognition 8.1 Fourier descriptors 8.2 Machine learning and deep learning for object recognition 8.3 Three-dimensional image analysis using deep learning	Weeks 14,15	CLO10, 10
Final exam	Week 16	

Lab season schedule		
Lab 1: Digital image representation	Week 1,	CLO2, CLO9
Lab 2: Image enhancement in the spatial domain	Week 2	CLO3, CLO9
Lab 3: Image restoration and reconstruction	Week 3,4	CLO5, CLO9
Lab 4: Image compression	Week 5	CLO6, CLO9
Lab 5: Color image processing	Week 6,7	CLO7, CLO9
Lab 6: Morphological image processing	Week 8	CLO7, CLO9
Lab 7: Image segmentation	Week 9,10	CLO8, CLO9
Lab 8: Image filtering	Week 11,	CLO3, CLO4, CLO9
Lab 9: Edge Detection	Week 12,13	CLO8, CLO9
Lab 10: Object Recognition	Week 14,15	CLO8, CLO10
Week 16, Final exam		
Methodology	The course will be delivered in lectures (with a participatory approach with particular attention to encourage students in asking questions and answering), present their assignment and actively participate in the lab session and project works	
Assessment Criteria	As per the academic regulation.	
Attendance	Lecture: 85%, Laboratory : 100%	
Course Policy	All students are expected to abide by the code of conduct of students (article 166 and 166.1.1, of The Senate Legislation of Bahir Dar University May 20, 2005) throughout this course. Academic dishonest including cheating, fabrication, and plagiarism will not be tolerated at any stage during your studies and will be reported to concerned bodies for action. If you need it you can get a copy (to be copied by yourself) of it from your academic advisor.	

Text Books and	1. R.J. Schalkoff, Wiley, Digital Image Processing and Computer Vision 2. R.C.Gonzalez & P.Wintz, Digital Image Processing - 2004 3. William K. Pratt, Digital Image Processing, 3rd edition, 2001
Reference book	1. Computer Vision - D.H.Ballard & C.M.Brown 2. Fundamentals of Digital Image Processing - A.K. Jain 3. Digital Picture Processing - A. Rosenfeld and A.C. Kak 4. Pattern Classification and Scene Analysis - R.O. Duda and P.E. Hart 5. Object Recognition by Computer - W.E.L. Grimson 6. Syntactic Pattern Recognition : An introduction -R.C.Gonzalez and M.G.Tho

Approval

The following bodies confirmed the course guidebook accordingly.

Position	Name	Signature	Date
Course Instructors	Alebachew Molla		03/06/2018
Course Module Chair	Dr.Gebeyehu Belay		
Chair Holder	Dr.Gebeyehu Belay		
V/Dean	Derejaw		